

1-An inviscid, steady, two-dimensional supersonic flow with freestream Mach number $M_\infty=3.0$, static pressure p_∞ and density ρ_∞ impinges on a slender, double-wedge body of chord length c . The first wedge surface is deflected by an angle $\alpha_1=5^\circ$ (up-wedge), and the second by $\alpha_2=2^\circ$ (down-wedge):

(a) Using the oblique-shock relation, find the shock angles that form on Surface 1 and Surface 2, and the corresponding downstream Mach numbers M_2 and M_3 .

(b) Compute the static pressures immediately behind each shock on both surfaces. From these, determine the pressure coefficient on each surface:

(c) Compute the drag coefficient and lift coefficient.

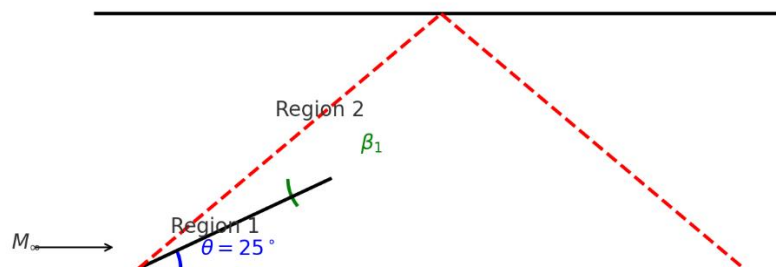
2- A two-dimensional inviscid flow at $M_\infty=3.3$, static temperature T_1 and pressure P_1 , encounters a flat plate that turns upward through an angle $\theta=25^\circ$. An oblique shock (Shock 1) forms at the ramp and impinges on a parallel ceiling above. Upon striking the ceiling, it reflects, producing a second oblique shock (Shock 2) that returns down toward the ramp, defining region 3 behind the reflected shock.

(a) Determine the oblique-shock angle at the ramp for Shock 1, and the downstream Mach number M_2 immediately behind it.

(b) Determine the reflected-shock angle and the Mach number M_3 in region 3 behind Shock 2.

(c) Compute the static temperature ratios across each shock:

(d) Compute the static pressure ratios:



4.16 Two compression ramps in supersonic flow create two plane oblique shocks, as shown. Calculate:

- (a) shock wave angle, β_1 (in degrees)
- (b) ramp angle, θ_1 (in degrees)
- (c) shock wave angle, β_2 (in degrees)
- (d) *estimate* the second ramp angle, θ_2 (in degrees).

